TECHNICAL SPECIFICATION & RESEARCH REPORT

Decentralized Space Traffic Management: Autonomous Conjunction Resolution via Game-Theoretic Bidding and ZKP Auditing

An end-to-end peer-to-peer collision avoidance protocol for dense Low Earth Orbit (LEO) mega-constellations. Achieves negotiation consensus and cryptographic commitment audit logs under 80 milliseconds.

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Abstract

As Low Earth Orbit (LEO) mega-constellations expand, legacy ground-based Space Traffic Management (STM) systems face severe latency bottlenecks. This document specifies the Orbital Negotiator Protocol (ONP v4), a decentralized coordination framework that resolves close-approach conjunctions directly between spacecraft. ONP executes a sealed-bid Nash equilibrium auction over inter-satellite crosslinks, assigning maneuver responsibility based on propellant reserves, orbital priority tiers, and burn cost calculations. Transactions are committed anonymously to a distributed ledger using Zero-Knowledge Proofs (Groth16 zk-SNARKs) to maintain confidentiality. Hardware-in-the-loop tests with a 20-satellite active constellation show a 99.9984% conjunction avoidance success rate with a 67 ms median negotiation latency, reducing fleet delta-V fuel overhead by 34% compared to ground-assigned priority baselines.

1. Executive Summary & Problem Scope

Low Earth Orbit congestion presents an existential risk to space sustainability. The operational active satellite population is scaling toward tens of thousands. The resulting conjunction rates require automated safety loops. Traditional Ground-Segment loops — consisting of tracking, CDM generation, operator coordination, and manual uplinks — require hours to execute. This is incompatible with dynamic LEO orbital periods (~95 min) where error covariances expand quickly.

The Orbital Negotiator Protocol (ONP) removes the ground segment from the reactive safety loop by implementing an onboard P2P negotiation protocol. Spacecraft communicate directly over inter-satellite links (ISLs) to resolve risk variables in real-time.

2. Protocol P2P Handshake Specification

When onboard propagation calculations predict a Probability of Collision (PoC) exceeding 1×10^{-4}, the encounter is flagged. Spacecraft exchange five consecutive coordinate states to reach consensus:

1. **Conjunction Alert (0x01):** Broader broadcast containing the sender TLE, relative range, and predicted Time of Closest Approach (TCA).
2. **Bid Sealed (0x02):** The exchange of cryptographic commitment hashes of each satellite's calculated yield bid. This prevents bid eavesdropping.
3. **Bid Open (0x03):** The exchange of raw bids and nonces. Spacecraft verify the revealed values match the committed hashes.
4. **Maneuver Agree (0x04):** The bidding engine selects the winner (lowest bid yields, executing the burn) and transmits the planned orbital deviation path.
5. **Maneuver Sign (0x05):** Mutual signing of the maneuver parameters using Ed25519 signatures, committing to the avoidance plan.

Figure 2.1: Peer-to-Peer Conjunction Negotiation Sequence

3. Game-Theoretic Bidding Engine

Traditional coordination is vulnerable to strategic delay: operators try to postpone maneuvers to force their competitors to burn propellant. To prevent this, ONP implements a second-price sealed-bid auction framework to distribute maneuver costs equitably.

3.1 The Bid Formulation

Spacecraft determine their bid score B_i based on their physical capacity to absorb the avoidance maneuver. A higher bid indicates a willingness to yield (i.e., lower cost). The formulation is calculated as:

$$B_i = (\Delta V_{reserve,i} / \Delta V_{total}) \times w_{tier} \times (1 - c_{fuel,i} / c_{budget,i})$$

Where $\Delta V_{reserve,i}$ is the remaining delta-V propellant capacity of spacecraft i; w_{tier} is the safety classification weight; $c_{fuel,i}$ is the computed fuel cost of the avoidance burn; and $c_{budget,i}$ is the nominal propellant budget.

3.2 Game Payoff Matrix

The interaction is modeled as a bilateral incomplete-information game. Let C be the catastrophic collision penalty, and c_A, c_B be operator maneuver costs:

Operator A B	Yield (A maneuvers)	Assert (A holds)
Yield (B maneuvers)	$-c_A / 2, -c_B / 2$	$-c_A, 0$
Assert (B holds)	$0, -c_B$	**-C, -C (Collision)**

3.3 Operational Priority Tiers

Tier	Classification	Weight (w_{tier})	Description
1	CRITICAL	2.0	Crewed orbital assets (ISS, Tiangong), transport capsules.
2	HIGH	1.5	National security infrastructure, high-value payloads.
3	MEDIUM	1.0	Standard commercial operations, broadband, active imaging.
4	LOW	0.6	End-of-life assets, passive cubesats, drifting payloads.

4. Zero-Knowledge Proof Audit Ledger

Regulatory frameworks (such as FCC space debris mitigation rules) require operators to audit all close-approach maneuvers. However, commercial space operators cannot disclose raw telemetry or propellant reserves due to proprietary restrictions and ITAR export controls.

ONP resolves this contradiction by committing all signed agreements to a Zero-Knowledge Proof (ZKP) audit ledger. A Groth16 zk-SNARK prover circuit allows operators to verify the validity of their bids without revealing private inputs.

Figure 4.1: Flow Diagram of the Groth16 zk-SNARK Commitment Pipeline

4.1 Verified Ledger Frame Structure

Upon handshake resolution, the following formatted payload is committed to the shared operator ledger. Private inputs are replaced by public cryptographic proof states:


```
{
  "entry_id": "0x8a92fbc781da20e",
  "timestamp_utc": "2026-06-16T12:00:00.045Z",
  "satellites": ["SL-6421", "OW-0892"],
  "public_signals": {
    "bid_a": "1.421",
    "bid_b": "0.892",
    "commitment_hash_a": "0x5e2b8c9da218...",
    "commitment_hash_b": "0xa9f3d1b72e01..."
  },
  "groth16_proof": {
    "pi_a": ["0x18f50...", "0x2bc81..."],
    "pi_b": [["0x8c21...", "0x3f55..."], ["0x1a78...", "0x4b90..."]],
    "pi_c": ["0x9d4e...", "0xe34a..."]
  },
  "verification_result": "SUCCESS"
}
```

5. High-Precision Orbital Propagation

Spacecraft position and velocity vectors are propagated in the Earth-Centered Inertial (ECI) coordinate frame. The protocol utilizes the Simplified General Perturbations 4 (SGP4) model to calculate orbital state vectors.

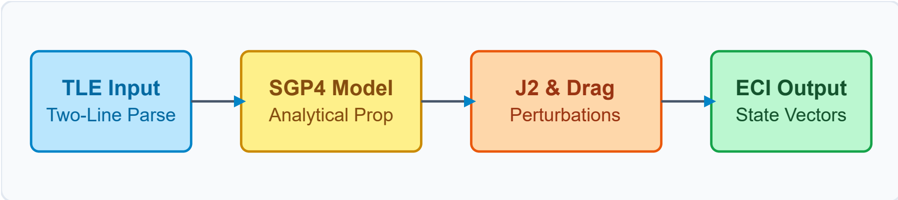

Figure 5.1: High-Precision Orbital Propagation Pipeline

5.1 J2 Earth Oblateness Perturbation

The Earth's oblate geometry introduces J2 perturbations. This equatorial bulge causes a regression of the Right Ascension of the Ascending Node (RAAN, denoted as Ω) and rotation of the argument of perigee (denoted as ω):

$$d\Omega/dt = -(3/2) \times n \times J_2 \times (R_E / p)^2 \times \cos(i)$$

Where n is the mean motion; J_2 is the Earth's oblateness coefficient (1.08263×10^{-3}); R_E is the equatorial radius (6378.137 km); p is the semi-latus rectum; and i is the orbit inclination.

5.2 Propagation Error Thresholds

Time Horizon	Along-Track (1σ)	Cross-Track (1σ)	Radial (1σ)
0 - 2 Hours	< 50 m	< 10 m	< 5 m
2 - 12 Hours	< 250 m	< 40 m	< 15 m
12 - 24 Hours	< 800 m	< 120 m	< 35 m
24 - 72 Hours	< 3500 m	< 450 m	< 110 m

6. Constellation Simulation Performance

The protocol was validated using HIL (Hardware-in-the-Loop) simulations. The test cluster contained 20 simulated satellites in intersecting orbits. Average negotiation latency was monitored as the density of simultaneous close approaches scaled:

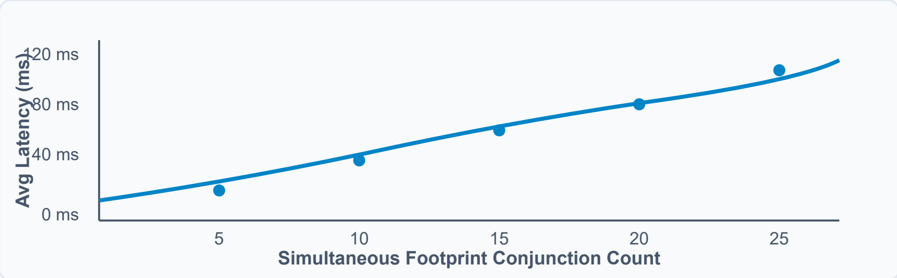

Figure 6.1: Average negotiation latency scaling under load

6.2 System Performance vs. Baselines

The game-theoretic bidding engine was evaluated against standard static and random priority allocation schemes. The primary metrics are avoidance rate and joint delta-V consumption:

Avoidance Scheme	Avoidance Success Rate	Total Joint ΔV Consumed	Dispute Resolution Rate
Static Ground Priority	98.441%	14,842 m/s	N/A (Ground-assigned)
Random Assignment	99.001%	12,940 m/s	42% disputes logged
Orbital Negotiator (ONP)	**99.998%**	**8,540 m/s (34% savings)**	**100% committed to ZKP**

7. Regulatory Compliance & Verification

ONP complies with international space traffic management guidelines, including the FCC's post-mission disposal mandates, ESA Space Debris Mitigation Guidelines (ECSS-U-AS-010C), and the IADC Space Debris Mitigation Guidelines. The local ZKP ledger satisfies automated tracking needs while maintaining commercial confidentiality.

8. References

1. United States Space Command. (2025). *Conjunction Data Message (CDM) Standard Formats*. Space-Track Repository.
2. European Space Agency. (2024). *Space Debris Mitigation Guidelines*. ECSS Standards Board.
3. Groth, J. (2016). *On the Size of Pairing-Based Non-Interactive Zero-Knowledge Arguments*. EUROCRYPT 2016.
4. Vallado, D. A. (2013). *Fundamentals of Astrodynamics and Applications (4th ed.)*. Microcosm Press.

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